

Honeycomb Serpentine: Satellite-Cluster Analysis Report

Updated with Cluster/Satellite Threshold & Detailed Analysis

1. Cluster vs Satellite Mode Definition

- Cluster: $\text{dot_count} \geq 4$ AND $\text{spread} \geq 200$ px
- Satellite: All other frames (low activity)

2. Summary Statistics by Video

NEW_175030: Dot mean=1.59, max=6.0, Spread mean=30.32, Cluster frames=0.0
NEW_175141: Dot mean=2.34, max=5.0, Spread mean=59.6, Cluster frames=0.0
NEW_175201: Dot mean=1.71, max=4.0, Spread mean=36.54, Cluster frames=0.0
NEW_175218: Dot mean=2.01, max=5.0, Spread mean=45.5, Cluster frames=0.0
NEW_175232: Dot mean=1.87, max=4.0, Spread mean=42.94, Cluster frames=0.0
OFF_171223: Dot mean=3.68, max=10.0, Spread mean=211.5, Cluster frames=117.0
OFF_171332: Dot mean=4.81, max=9.0, Spread mean=272.67, Cluster frames=153.0
OFF_171702: Dot mean=10.79, max=16.0, Spread mean=303.83, Cluster frames=201.0
OFF_171802: Dot mean=9.19, max=13.0, Spread mean=303.68, Cluster frames=163.0
OFF_171833: Dot mean=11.02, max=16.0, Spread mean=297.69, Cluster frames=230.0

3. State Distribution & Transitions

OFF_171223: 117 Cluster | 116 Satellite

OFF_171332: 153 Cluster | 63 Satellite

OFF_171702: 201 Cluster | 27 Satellite

OFF_171802: 163 Cluster | 38 Satellite

OFF_171833: 230 Cluster | 32 Satellite

NEW_175030: 0 Cluster | 218 Satellite

NEW_175141: 0 Cluster | 239 Satellite

NEW_175201: 0 Cluster | 368 Satellite

NEW_175218: 0 Cluster | 292 Satellite

NEW_175232: 0 Cluster | 335 Satellite

OFF conditions show frequent Cluster ↔ Satellite switching.

NEW conditions predominantly Satellite mode.

4. Per-Video Time Series & Next Steps

Detailed dot_count + spread overlay plots generated separately.

time_s used as rotation angle proxy for transition analysis.

Carbon coating comparison (θ_{C} vs θ_{S}) planned for future experiments.

Files available:

- Updated report (this PDF)
- detailed_time_series_plots.pdf
- updated_analysis_with_states.csv